

SEVENSTONES LIGHTSHIP 62107, United Kingdom

WMO: 992700

Lat: 50.102N

Lon: 6.100W

Elev: 0

StdP: 101.33

Time Zone: 0.00 (EUW)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.9	4.3	-4.9	2.5	7.0	-3.5	2.8	6.6	22.8	8.6	21.4	8.4	9.6	60	0.899

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.5	18.5	16.9	17.9	16.4	17.5	16.0	17.5	18.1	16.9	17.5	16.4	17.1	6.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.2	17.8	16.6	11.8	17.3	16.1	11.4	16.9	49.1	18.2	47.4	17.6	46.1	17.2	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	18.3	16.3	DB	1.5	19.4	1.9	1.5	0.1	20.5	-1.0	21.4	-2.1	22.3	-3.5	23.4	(4)
(5)			WB	-0.4	18.0	1.8	1.2	-1.7	18.8	-2.7	19.5	-3.7	20.2	-5.0	21.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	9.0	8.2	8.5	9.7	11.4	13.9	15.7	16.1	15.3	13.6	11.4	9.8
	DBStd	3.16	1.93	1.90	1.72	1.27	1.15	1.25	1.27	0.87	1.05	1.38	1.83	1.92
	HDD10.0	193	39	52	48	18	2	0	0	0	0	1	8	26
	HDD18.3	2342	288	283	303	258	214	134	82	70	92	147	207	263
	CDD10.0	894	9	2	3	10	46	116	178	188	158	112	51	21
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.8	10.9	10.2	9.3	8.2	8.1	7.2	7.2	7.3	7.7	8.7	10.1	11.0
Precipitation	PrecAvg	1160	128	97	83	76	70	74	76	86	85	126	131	128
	PrecMax	1445	237	204	161	148	182	171	207	247	176	223	238	268
	PrecMin	972	34	25	21	24	5	20	18	17	20	50	49	42
	PrecStd	124	48	47	36	34	34	39	38	47	39	47	50	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	12.2	11.7	13.0	15.3	17.5	20.1	19.0	18.2	16.7	14.8	13.9
		MCWB	11.6	10.7	10.2	10.7	12.7	15.6	18.1	17.4	16.5	15.7	13.6	12.6
	2%	DB	12.3	11.5	11.2	12.1	14.0	16.7	18.9	18.3	17.6	16.1	14.4	13.4
		MCWB	11.1	9.8	9.9	10.7	12.4	15.1	17.2	16.7	16.2	15.0	13.2	12.2
	5%	DB	11.8	10.9	10.8	11.7	13.5	16.2	18.1	17.9	17.2	15.6	14.0	12.8
		MCWB	10.6	9.0	9.6	10.4	12.1	14.6	16.6	16.4	15.8	14.5	12.9	11.6
	10%	DB	11.4	10.5	10.5	11.3	13.0	15.6	17.5	17.5	16.7	15.2	13.6	12.3
		MCWB	10.1	8.5	9.3	10.1	11.9	14.1	16.2	16.0	15.4	14.0	12.3	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.3	11.1	11.9	13.4	16.2	18.5	18.0	17.2	16.2	14.2	13.2
		MCDB	12.5	11.7	11.3	12.1	14.1	16.9	19.3	18.4	17.5	16.5	14.5	13.6
	2%	WB	11.6	10.6	10.5	11.2	12.9	15.6	17.6	17.5	16.7	15.5	13.6	12.5
		MCDB	12.1	11.2	10.9	11.7	13.5	16.4	18.5	17.9	17.2	15.9	14.1	13.1
	5%	WB	11.1	10.0	10.0	10.8	12.5	15.1	17.0	16.9	16.3	15.0	13.1	11.9
		MCDB	11.6	10.7	10.6	11.4	13.2	15.9	17.8	17.5	16.8	15.4	13.8	12.7
	10%	WB	10.5	9.4	9.6	10.4	12.1	14.5	16.4	16.4	15.8	14.6	12.6	11.3
		MCDB	11.3	10.3	10.3	11.1	12.8	15.4	17.3	17.1	16.5	15.1	13.5	12.2
Mean Daily Temperature Range	5% DB	MDBR	2.3	2.1	1.9	1.6	1.5	1.5	1.5	1.5	1.5	1.7	2.2	2.3
		MCDBR	2.4	2.2	1.9	1.8	2.0	1.9	2.0	1.8	1.7	1.8	2.2	2.6
		MCWBR	3.2	2.7	2.4	2.0	1.8	1.9	1.9	2.1	2.1	2.5	3.1	3.5
	5% WB	MCDBR	2.2	2.3	1.8	1.6	1.9	1.8	1.7	1.8	1.7	1.8	2.2	2.5
MCWBR		3.3	3.0	2.4	2.0	2.0	2.0	2.0	2.3	2.2	2.7	3.2	3.6	
Clear-Sky Solar Irradiance	taub	0.344	0.356	0.384	0.386	0.395	0.381	0.388	0.371	0.369	0.364	0.342	0.336	
	taud	2.339	2.370	2.320	2.351	2.374	2.449	2.437	2.498	2.486	2.442	2.386	2.307	
	Ebn at Noon	677	774	815	857	863	878	865	866	828	757	676	619	
	Edh at Noon	69	85	105	113	115	108	108	97	89	78	65	62	
All-Sky Solar Radiation	RadAvg	0.86	1.65	2.79	4.58	5.50	6.04	5.49	4.72	3.55	2.07	1.10	0.69	
	RadStd	0.07	0.14	0.30	0.28	0.40	0.51	0.41	0.30	0.24	0.12	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page